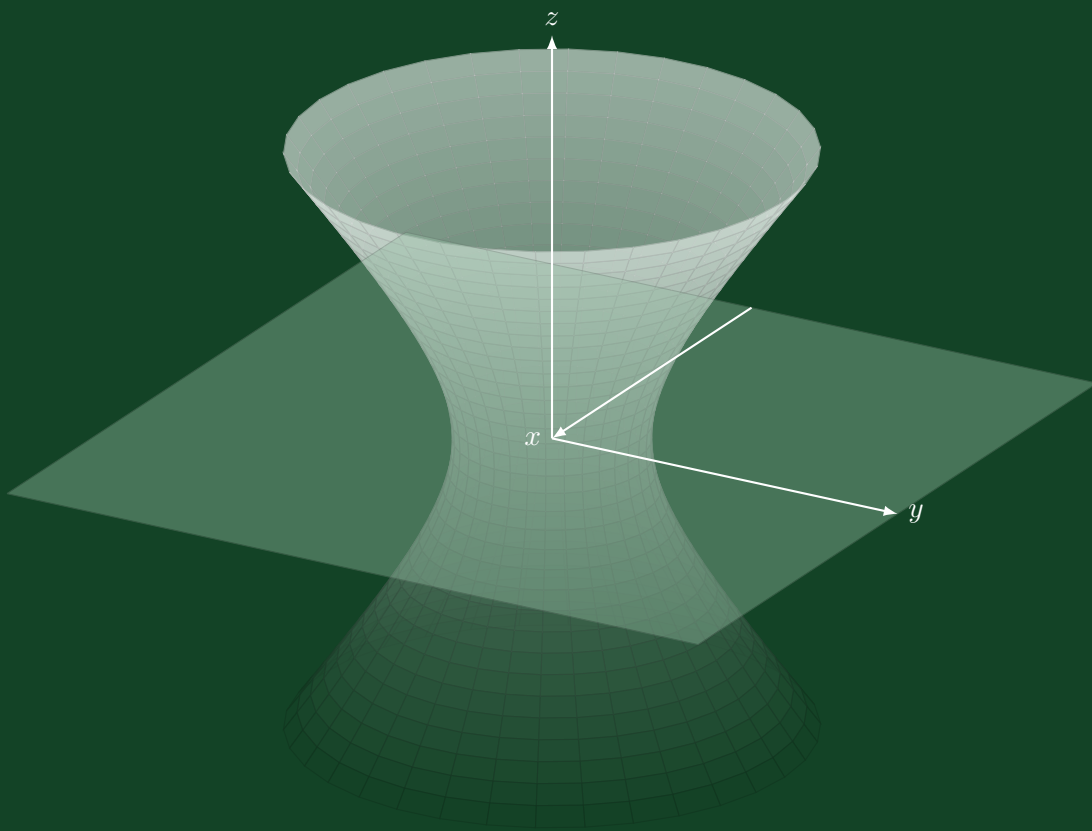


DISCRETE PROBABILITY DISTRIBUTIONS

An Intuitive & Mathematical Approach

ASSIGNMENT 4 SOLUTIONS



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1 The Geometric Distribution

Question 1

Derive the probability mass function (PMF) of the geometric distribution.

Solution:

Consider a sequence of independent Bernoulli trials, each with a constant probability of success p and a probability of failure $q = 1 - p$. Let the discrete random variable X denote the number of failures preceding the first successful trial.

If exactly $(x + 1)$ trials are required to achieve the first success, then the first x trials must be failures, and the $(x + 1)$ -th trial must be a success. The random variable X can take on non-negative integer values: $x = 0, 1, 2, \dots, \infty$.

Because the trials are independent, the joint probability of x failures followed by one success is the product of their individual probabilities. We observe the following sequence of probabilities:

$$\begin{aligned}P(X = 0) &= p \\P(X = 1) &= qp \\P(X = 2) &= q^2p \\&\vdots \\P(X = x) &= q^x p\end{aligned}$$

Thus, the probability mass function (PMF) of the geometric distribution is given by:

$$f(x; p) = P(X = x) = pq^x, \quad \text{for } x = 0, 1, 2, \dots \quad (1)$$

Question 2

Show that the probability mass function (PMF) of the geometric distribution satisfies the axioms of probability.

Solution:

To be a valid probability mass function, $f(x)$ must satisfy two fundamental axioms:

Axiom 1: Non-negativity

Since p is a probability, $0 < p < 1$, which implies $0 < q < 1$. Consequently, for any non-negative integer x , $q^x \geq 0$. Therefore:

$$P(X = x) = pq^x \geq 0 \quad \text{for all } x \in \{0, 1, 2, \dots\}$$

Axiom 2: Total Probability Equals One

We must show that the sum of all probabilities over the support of X is exactly 1.

$$\begin{aligned}\sum_{x=0}^{\infty} P(X = x) &= \sum_{x=0}^{\infty} pq^x \\&= p \sum_{x=0}^{\infty} q^x \\&= p(1 + q + q^2 + q^3 + \dots)\end{aligned}$$

Recognizing the infinite geometric series with a common ratio $q \in (0, 1)$, its sum is $\frac{1}{1-q}$. Thus:

$$\sum_{x=0}^{\infty} P(X = x) = p \left(\frac{1}{1-q} \right)$$

Since $p + q = 1$, we have $1 - q = p$. Substituting this yields:

$$\sum_{x=0}^{\infty} P(X = x) = p \left(\frac{1}{p} \right) = 1$$

Therefore, $f(x) = pq^x$ is a valid probability mass function.

Question 3

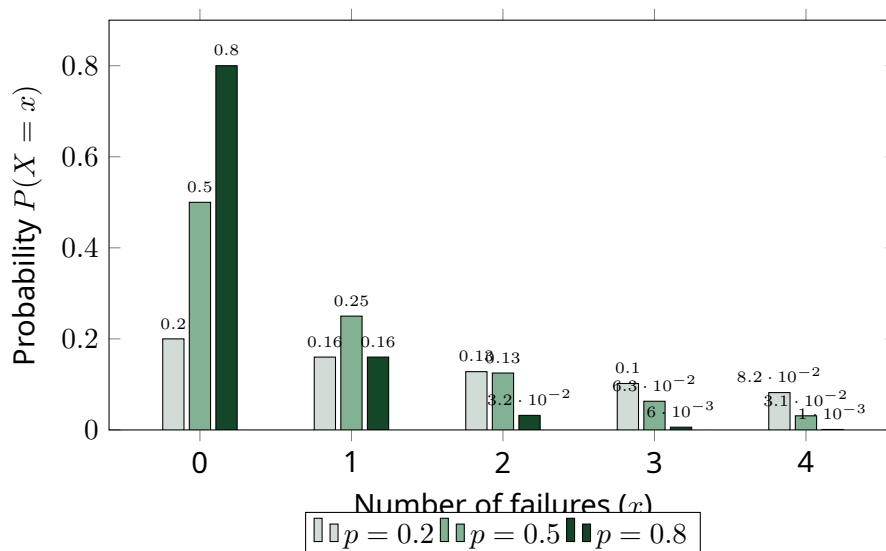
Explain, with examples, how the probability mass function (PMF) of the geometric distribution changes as the probability of success increases.

Solution:

As the probability of success (p) increases, the geometric distribution becomes more heavily right-skewed and concentrated near $x = 0$. This intuitively indicates that with a higher success rate, the expected number of failures before the first success rapidly decreases.

Consider three scenarios with varying probabilities of success:

- **Small Probability** ($p = 0.2, q = 0.8$): The PMF decreases slowly, leaving a long right tail.
- **Moderate Probability** ($p = 0.5, q = 0.5$): The PMF decreases at a moderate exponential rate.
- **Large Probability** ($p = 0.8, q = 0.2$): The PMF decreases rapidly and is highly concentrated at $x = 0$.



As visually demonstrated, increasing the probability of success shifts the statistical weight toward the origin, confirming that the first success is expected to occur sooner.

Question 4

Derive the moment generating function (MGF) of the geometric distribution and hence obtain its coefficient of variation.

Solution:

The moment generating function (MGF), $M_X(t)$, is defined as the expected value of e^{tX} :

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] = \sum_{x=0}^{\infty} e^{tx} P(X = x) \\ &= \sum_{x=0}^{\infty} e^{tx} (pq^x) \\ &= p \sum_{x=0}^{\infty} (qe^t)^x \end{aligned}$$

This is an infinite geometric series with common ratio (qe^t) . For convergence, we require $qe^t < 1$, which means $t < -\ln(q)$. Summing the series yields the MGF:

$$M_X(t) = \frac{p}{1 - qe^t} \quad (2)$$

Mean (μ): We evaluate the first derivative of the MGF at $t = 0$.

$$\begin{aligned} M'_X(t) &= \frac{d}{dt} (p(1 - qe^t)^{-1}) = -p(1 - qe^t)^{-2}(-qe^t) = \frac{pqe^t}{(1 - qe^t)^2} \\ \mu &= \mathbb{E}[X] = M'_X(0) = \frac{pq}{(1 - q)^2} = \frac{pq}{p^2} = \frac{q}{p} \end{aligned}$$

Variance (σ^2): We evaluate the second derivative at $t = 0$.

$$\begin{aligned} M''_X(t) &= \frac{d}{dt} \left(\frac{pqe^t}{(1 - qe^t)^2} \right) = \frac{pqe^t}{(1 - qe^t)^2} + \frac{2pq^2e^{2t}}{(1 - qe^t)^3} \\ \mathbb{E}[X^2] &= M''_X(0) = \frac{pq}{(1 - q)^2} + \frac{2pq^2}{(1 - q)^3} = \frac{q}{p} + \frac{2q^2}{p^2} = \frac{q(p + 2q)}{p^2} = \frac{q(1 + q)}{p^2} \end{aligned}$$

Using the variance formula $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$:

$$\text{Var}[X] = \frac{q(1 + q)}{p^2} - \frac{q^2}{p^2} = \frac{q + q^2 - q^2}{p^2} = \frac{q}{p^2}$$

Standard Deviation: $\sigma = \sqrt{\text{Var}[X]} = \frac{\sqrt{q}}{p}$.

Coefficient of Variation (CV):

$$\text{CV} = \left(\frac{\sigma}{\mu} \right) \times 100\% = \frac{\frac{\sqrt{q}}{p}}{\frac{q}{p}} \times 100\% = \frac{1}{\sqrt{q}} \times 100\%$$

Question 5

State and prove the memoryless property of the geometric distribution.

Solution:

A discrete random variable X possesses the **memoryless property** if, for any non-negative integers m and n , it satisfies:

$$P(X \geq m + n \mid X \geq m) = P(X \geq n)$$

This signifies that the probability of waiting for n additional failures does not depend on how many failures (m) have already occurred. The distribution "forgets" the past.

Proof:

First, we find the cumulative survival probability $P(X \geq k)$:

$$P(X \geq k) = \sum_{x=k}^{\infty} pq^x = pq^k \sum_{i=0}^{\infty} q^i = pq^k \left(\frac{1}{1-q} \right) = \frac{pq^k}{p} = q^k$$

Using the definition of conditional probability:

$$\begin{aligned} P(X \geq m + n \mid X \geq m) &= \frac{P((X \geq m + n) \cap (X \geq m))}{P(X \geq m)} \\ &= \frac{P(X \geq m + n)}{P(X \geq m)} \quad (\text{since } X \geq m + n \subseteq X \geq m) \\ &= \frac{q^{m+n}}{q^m} \\ &= q^n \\ &= P(X \geq n) \end{aligned}$$

Thus, the memoryless property is proven.

2 The Negative Binomial Distribution

Question 6

Derive the probability mass function (PMF) of the negative binomial distribution, and explain why it is called the "negative binomial" distribution.

Solution:

Suppose we have a succession of independent Bernoulli trials where the probability of success, p , remains constant from trial to trial. Let $f(x; n, p)$ denote the probability that there are exactly x failures preceding the n -th success.

For the n -th success to occur on the $(x + n)$ -th trial, two independent conditions must be met:

1. There must be exactly $(n - 1)$ successes and x failures among the first $(x + n - 1)$ trials. The probability of this arrangement is given by the binomial probability:

$$\binom{x + n - 1}{n - 1} p^{n-1} q^x$$

2. The final, $(x + n)$ -th trial must be a success, which has a probability of p .

Multiplying these probabilities together yields the PMF:

$$f(x; n, p) = \binom{x + n - 1}{n - 1} p^{n-1} q^x \cdot p = \binom{x + n - 1}{n - 1} p^n q^x, \quad x = 0, 1, 2, \dots \quad (3)$$

Nomenclature:

Unlike the ordinary binomial distribution where the number of trials is fixed and the number of successes is random, the negative binomial distribution fixes the number of successes (n) and allows the number of trials (or failures, x) to be random. Furthermore, it derives its name from the fact that the probabilities represent successive terms in the binomial expansion of $(1 - q)^{-n}$ (which involves a negative exponent).

Question 7

Derive the moment generating function (MGF) of the negative binomial distribution and hence obtain its coefficient of variation.

Solution:

The MGF is calculated as follows:

$$\begin{aligned} M_X(t) &= \mathbb{E}[e^{tX}] = \sum_{x=0}^{\infty} \binom{x + n - 1}{x} p^n q^x e^{tx} \\ &= p^n \sum_{x=0}^{\infty} \binom{x + n - 1}{x} (qe^t)^x \end{aligned}$$

Using the negative binomial expansion identity $\sum_{x=0}^{\infty} \binom{x + n - 1}{x} z^x = (1 - z)^{-n}$, we substitute $z = qe^t$:

$$M_X(t) = p^n (1 - qe^t)^{-n} = \left(\frac{p}{1 - qe^t} \right)^n \quad (4)$$

By utilizing the logarithmic derivative method ($\ln M_X(t) = n \ln p - n \ln(1 - qe^t)$) and differentiating, we obtain the moments:

$$\mu = \mathbb{E}[X] = M'_X(0) = \frac{nq}{p}$$

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{nq}{p^2}$$

Standard Deviation: $\sigma = \sqrt{\text{Var}[X]} = \frac{\sqrt{nq}}{p}$.

Coefficient of Variation (CV):

$$\text{CV} = \frac{\sigma}{\mu} \times 100\% = \frac{\sqrt{nq}/p}{nq/p} \times 100\% = \frac{1}{\sqrt{nq}} \times 100\%$$

Question 8

Show that the sum of independent negative binomial random variables having a common parameter p is also a negative binomial random variable.

Solution:

Let $X_1 \sim \text{NB}(n_1, p)$ and $X_2 \sim \text{NB}(n_2, p)$ be independent random variables. The MGF of a sum of independent random variables is the product of their individual MGFs:

$$\begin{aligned} M_{X_1+X_2}(t) &= M_{X_1}(t) \cdot M_{X_2}(t) \\ &= \left(\frac{p}{1 - qe^t} \right)^{n_1} \cdot \left(\frac{p}{1 - qe^t} \right)^{n_2} \\ &= \left(\frac{p}{1 - qe^t} \right)^{n_1+n_2} \end{aligned}$$

This resulting function is exactly the MGF of a negative binomial distribution with parameters $(n_1 + n_2, p)$. Therefore, $X_1 + X_2 \sim \text{NB}(n_1 + n_2, p)$.

Question 9

Show that, under suitable limiting conditions, the negative binomial distribution converges to the Poisson distribution.

Solution:

The PMF of the negative binomial is $P(X = x) = \frac{n(n+1)\dots(n+x-1)}{x!} p^n q^x$.

Let $n \rightarrow \infty$ and $q \rightarrow 0$ such that the mean $nq = \lambda$ (a constant). Thus, $q = \frac{\lambda}{n}$ and $p = 1 - \frac{\lambda}{n}$. We substitute these into the PMF:

$$\begin{aligned} \lim_{n \rightarrow \infty} P(X = x) &= \lim_{n \rightarrow \infty} \frac{n(n+1)\dots(n+x-1)}{x!} \left(1 - \frac{\lambda}{n}\right)^n \left(\frac{\lambda}{n}\right)^x \\ &= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} \left[\frac{n}{n} \cdot \frac{n+1}{n} \dots \frac{n+x-1}{n} \right] \left(1 - \frac{\lambda}{n}\right)^n \end{aligned}$$

As $n \rightarrow \infty$: 1. The fractional terms $(1 + \frac{1}{n}) \dots (1 + \frac{x-1}{n}) \rightarrow 1$. 2. The term $(1 - \frac{\lambda}{n})^n \rightarrow e^{-\lambda}$. Therefore:

$$\lim_{n \rightarrow \infty} P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

This is exactly the PMF of the Poisson distribution.

3 Relationships Between Distributions

Questions 10 & 11

Explain why the geometric and negative binomial distributions are regarded as the discrete counterparts of the exponential and gamma distributions, respectively.

Solution:

These distribution pairs represent fundamentally identical physical processes modeled in different mathematical spaces (discrete trials vs. continuous time).

Geometric Distribution (Discrete)	Exponential Distribution (Continuous)
Models waiting time in <i>number of trials</i> .	Models waiting time in <i>continuous time</i> .
Discrete random variable.	Continuous random variable.
Counts failures before the first success.	Measures time until the first event.
Possesses the memoryless property.	Possesses the memoryless property.
PMF: $P(X = x) = pq^x$	PDF: $f(x) = \lambda e^{-\lambda x}$

Similarly, just as the Negative Binomial distribution is the sum of n independent Geometric random variables, the Gamma distribution is the sum of n independent Exponential random variables.

Negative Binomial Distribution	Gamma Distribution
Waiting time in <i>number of trials</i> .	Waiting time in <i>continuous time</i> .
Counts failures before the n-th success.	Measures time until the n-th event.
Sum of n i.i.d. geometric variables.	Sum of n i.i.d. exponential variables.

Questions 12 & 13

Derive the logarithmic series distribution from both the geometric series and as a limiting form of the negative binomial distribution.

Solution:

1. From the Geometric Series:

The geometric series expansion is $\frac{1}{1-p} = 1 + p + p^2 + p^3 + \dots$

Integrating both sides with respect to p yields:

$$-\ln(1-p) = p + \frac{p^2}{2} + \frac{p^3}{3} + \dots$$

To create a valid probability distribution (where probabilities sum to 1), we divide the series by its sum, $-\ln(1-p)$:

$$1 = \frac{p}{-\ln(1-p)} + \frac{p^2}{-2\ln(1-p)} + \frac{p^3}{-3\ln(1-p)} + \dots$$

This yields the Logarithmic Series PMF: $P(X = x) = \frac{p^x}{-x\ln(1-p)}$ for $x = 1, 2, 3, \dots$

2. As a Limiting Form of Negative Binomial:

Given $X \sim \text{NB}(n, p)$, its PMF is $P(X = x) = \frac{\Gamma(n+x)}{x!\Gamma(n)} p^n q^x$. We observe the conditional probability of X given $X > 0$:

$$P(X = x \mid X > 0) = \frac{P(X = x)}{1 - P(X = 0)} = \frac{\frac{\Gamma(n+x)}{x!\Gamma(n)} p^n q^x}{1 - p^n}$$

Taking the limit as $n \rightarrow 0$, utilizing L'Hôpital's rule on the ratio of approaching zero terms:

$$\lim_{n \rightarrow 0} P(X = x \mid X > 0) = \frac{q^x}{-x \ln(1 - q)}$$

Substituting parameters yields the standard logarithmic distribution formulation.

4 Data-Driven Model Identification

Question 14

Describe how the observed sample mean and sample variance can be used to identify the most appropriate discrete probability distribution.

Solution:

When modeling discrete data, comparing the sample mean ($\bar{x}$) and sample variance (s^2) is a critical diagnostic tool. Different distributions exhibit distinct theoretical mean-to-variance ratios (dispersion indices).

By calculating $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ and $s^2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}$, one can map the observed data properties to the theoretical properties summarized below:

Distribution	Mean (μ)	Variance (σ^2)	Identification Criterion
Binomial	np	npq	$s^2 < \bar{x}$; Under-dispersed data with a strict, finite upper limit.
Poisson	λ	λ	$s^2 \approx \bar{x}$; Equi-dispersed data modeling rare events.
Geometric	$\frac{q}{p}$	$\frac{q}{p^2}$	$s^2 > \bar{x}$; Values start at 0, frequencies decrease strictly and rapidly.
Negative Binomial	$\frac{nq}{p}$	$\frac{nq}{p^2}$	$s^2 > \bar{x}$; Over-dispersed count data.
Logarithmic	$\frac{-p}{(1-p) \ln(1-p)}$	$\frac{-p(p + \ln(1-p))}{(1-p)^2 [\ln(1-p)]^2}$	$s^2 > \bar{x}$; No zero observations, massive spike at $x = 1$, long tail.